MILITARY SPECIFICATION

TRUCK TRACTOR, ON-OFF ROAD, ENGINEER

MIN. 120,000 GCWR,

COMMERCIAL CONSTRUCTION EQUIPMENT

(CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an on-off road, heavy-duty hauler, commercial truck-tractor in an 8 x 4 articulated combination vehicle configuration, having at least a 69,000 GVWR and 120,000 GCWR, sustain at least a 47,000 pound vertical kingpin weight load, and features permitting heavy hauling on construction jobs. The truck tractor is one version of the family of Commercial Construction Equipment (CCE) for engineer troop use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-S-40626	- Sign Kit, Vehicle Class.

FSC 3805

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MIL-C-52895 - Chassis-Cab, Multi-Axle, On-Off-Road, Commercial Construction, Equipment (CCE), General Specification for.

STANDARDS

FEDERAL

FED. STD. No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z 35.1 - Industrial Accident Prevention Signs, Specifications for.
ANSI B 93.31 - Multi-Pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Book of ASTM Standards - Parts 1 through 11.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

DEPARTMENT OF LABOR (DOL)

29 CFR 1926 - Occupational Safety and Health Administration Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 567 - Certification.
49 CFR 571 - Federal Motor Vehicle Safety Standards (FMVSS).
49 CFR 393 - Federal Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC., (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, N.W., Washington, DC 20005.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, State of California, 2570 24th Street, Sacramento, CA 95804.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

TTMA Recommended Practices.
TTMA Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., Suite 203, 2430 Pennsylvania Avenue, N. W., Washington, DC 20037.)

3. REQUIREMENTS

3.1 Description. The truck tractor shall be as specified herein and in the chassis-cab specification, MIL-C-52895(ME). The truck tractor shall include headache rack, hydraulic winch, slider lower coupler (fifth wheel), and chassis-cab assemblies. The truck tractor shall be a commercial, on-off road, 4-axle, diesel-engine-driven type for severe construction applications. The truck tractor shall be the latest commercial production model of the contractor and shall have been marketed for a period of not less than 1 year prior to opening of the technical proposal or bid, except for product improvements which have been incorporated on commercial units. The truck tractor shall be equipped with

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all components furnished as standard and optional equipment, whether specified herein or not, that are necessary to enable the truck tractor to function safely, reliably and efficiently in severe, sustained construction job operations. Established component and vehicle ratings and capacities shall not be increased to meet the requirements of this specification.

3.1.1 Environmental conditions. The truck tractor shall start, travel, and function as specified herein under world-wide climatic conditions from minus 25°F to plus 120°F and from sea level to a minimum 5000 foot altitude.

3.2 Truck tractor equipment construction.

3.2.1 Truck tractor assembly. The truck tractor equipment shall be of the heavy-duty types intended for construction applications in severe usage world-wide in heavy hauler service. The truck tractor shall include headache rack and winch, slider type lower coupler (fifth wheel) and deck plates, and hydraulic system, all mounted on an on-off road chassis-cab.

3.2.2 Performance. The truck tractor equipment, as installed on the chassis-cab, shall comply with the interrelated requirements as follows:

(a) Perform hauling cycles of coupling to a loaded semitrailer, transport to the drop site, uncouple from the semitrailer, and return to the pick-up site, both bobtail (without a semitrailer) and towing an empty semitrailer.

(1) Perform short hauling cycles of under 3 miles for at least 10 hours;
(2) Perform long hauling cycles of over 400 miles on highway;
(3) Transport semitrailers, up to GCWR, on express ways, primary highways, unpaved secondary roads, construction haul roads, and off-road mud, snow, and water crossings up to the limits of speed or traction;
(4) Couple and uncouple with semitrailers on off-road terrain normally encountered at earthmoving construction job sites.

(b) Be compatible with the following semitrailer equipment;

(1) Heavy hauler machinery semitrailers.
(2) Tanker semitrailers.
(3) Platform semitrailers.

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(4) Van and shop semitrailers.
(5) Asphalt plants.
(6) Cement plants.
(7) Rock crushing and screening plants.
(8) Sawmills.
(9) Drilling rigs.
(10) Self-loader, freight container semitrailers.

3.2.3 Headache rack and winch equipment. An oil field type headache rack and winch shall be furnished between the cab and the lower coupler as follows:

(a) Provide headache rack fabricated of steel equal to ASTM 120, minimum 4-inch diameter, Schedule 60 pipe and covered with expanded metal; comply with DOT FMCSR for use with low bed machinery semi-trailers when transporting pipe, timbers, and poles.
(1) Minimum 4 each vertical members; 2 each frame width and 2 each cab width.
(2) Minimum 4 each horizontal members; 1 each top of cab height and 1 each top of frame and 2 each intermediate.
(3) Twin, angled longitudinal braces from headache rack rearward to frame with winch cradled between the braces.
(4) Near top of rack, minimum 12 each hooks for hanging and securing come-a-longs, logging chains, and tire chains; hinged angle iron keeper(s) covering hook ends and latches to prevent loss.
(5) All open pipe ends capped and all joints welded watertight.
(6) Installation to chassis-cab, same as for lower coupler assembly installation, comply with DOT semitrailer payload forces, compatible with winch pulling forces, and approved by truck manufacturer.

(b) Provide two-speed hydraulic driven, planetary drum winch having a minimum of 45,000 pound line pull with one layer of cable on the drum and at hydraulic relief valve pressure setting.

(1) Winch complete with cable fairlead rolls, horizontal and vertical, and line tension roller; fairlead located to result in winch cable clearing the truck tractor coupler plate by minimum 2 inches.
(2) Duplicate winch controls; in-cab and road side ground locations, deadman types.

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(3) Controls to provide "Winch-in", "Hold", and "Winch-out" functions.
(4) Provide positive mechanical neutral control lock, holding both sets of controls in "Hold"; disengagement of either neutral control lock to permit operator to activate the winch.
(5) Provide winch-in cable speeds, at one layer of cable on drum, of maximum 8 fpm at low speed and 2-1/2 times the low speed maximum at high speed.
(6) Provide winch safety brake of the spring applied hydraulic pressure release type.
(7) Installation of winch in accordance with both winch manufacturer and chassis manufacturer for construction applications.

(c) Provide minimum of 100 feet of wire rope on winch drum:

(1) Minimum 7/8-inch diameter size, 6 by 31, IWRC, very high strength (VHS) winch wire rope having each wire strand individually formed providing maximum exterior wear surface area.
(2) Working end terminate in swaged type pendent fitting, swivel, minimum 4 foot length of chain, and safety swivel hook.

3.2.4 Work lamps. The headache racks of the truck tractor shall be equipped with four each sealed beam work lamps in accordance with Figure 1 and as follows:

(a) Provide SAE J598 seal beam unit, PAR 46, Trade No. 4478 installed in shock mount rubber ring construction equipment type lamp housing.
(b) Mount lamp housing on the headache rack:

(1) Install 2 each lamps on curbside and 2 each lamps on road side, as far apart as possible and as high as practicable without any part of these lamps being outside of the headache rack perimeter;
(2) Mount lamps in accordance with SAE J598;
(3) Aim the two inboard or lower lamps to illuminate the rearward working area; these lamps may also function as automatic back-up lamps of MIL-C-52895;
(4) Aim the two outboard or upper lamps to illuminate the side working area, curbside and roadside of the truck tractor.

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(c) Provide SAE J553 circuit breakers, minimum 14 GA SAE J878 type HTS wire, and SAE J561 spade terminals; provide labled floodlamps switches on the instrument panel near the main lighting switch; minimum 20-amp rated capacity per switch; one each for side work areas and one each for rearward work area; provide adjacent amber indicator lamp for each switch.

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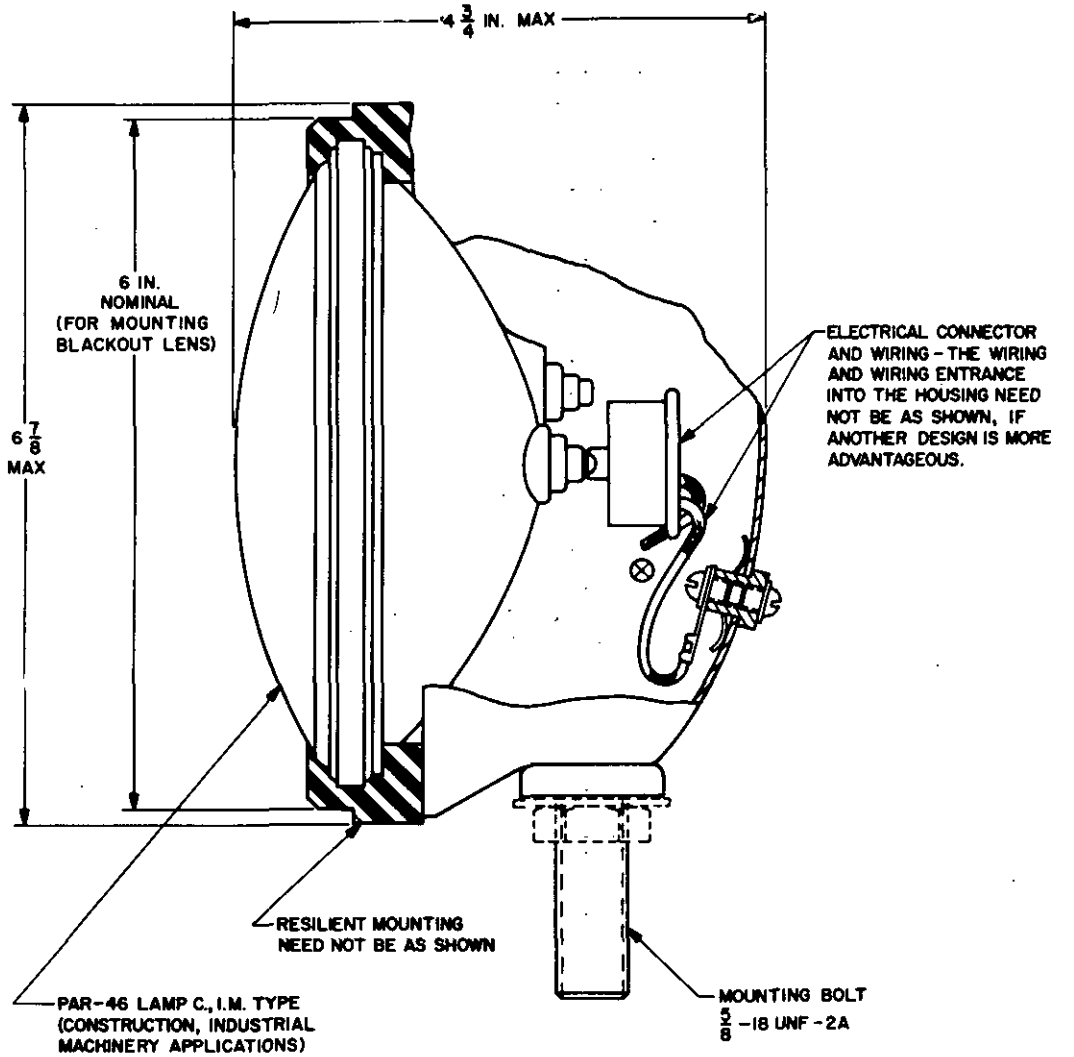


FIGURE 1.
HOUSING, LAMP, RESILIENT MOUNTING
DX-250

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3.2.5 Lower coupler equipment. The lower coupler equipment shall be in accordance with DOT FMCSR, SAE J701, and TTMA RP No. 8. The equipment shall include fifth wheel coupler, coupler adapter, slider, ramps, tail roller, and deck plates as follows:

(a) Provide lower coupler, minimum 36-inch diameter size, of the fore and aft rocking type having rated load capacities of:

(1) Minimum 50,000 pound vertical weight load rating.
(2) Minimum 15,000 pound vertical lifting reaction force.
(3) Minimum 60,000 pound horizontal dynamic loads.
(4) Accept an SAE J700, 2-inch throat diameter kingpin.
(5) Provide hardened steel jaws, fully encircling type, with slack adjustment.
(6) Provide air operated jaw release having driver air control on road side rear of cab accessible from the ground.

(b) Provide lower coupler dimensions (without adapter installed) in accordance with SAE J701:

(1) Forward Clearance (CFW) - Minimum 64-inches with FW settings of 56 inches or less.
(2) Coupler setting (FW) - Range of at least 14-inches through 56-inches from rearward to forward FW settings.
(3) Coupler Plate Height - 53-inches nominal.

(c) Provide lower coupler adapter as follows:

(1) Provide nominal 10-inch height increase to:
Raise lower coupler from 53-inches nominal to 63-inches nominal;
Change compatability with 2.0-inch SAE J700 kingpin to 3.5-inch SAE J848 kingpin;
(2) Upper portion of adapter include a 36-inch diameter coupler plate with 3.5-inch jaws and manual release mechanism.
(3) Lower portion of adapter include a reinforced minimum 1/4-inch thick upper coupler plate equipped with a heat treated, alloy steel 2.0-inch SAE J700 kingpin and triangular shaped 3/4-inch thick steel stops, nestled in between the truck tractor plate forks; result in the adapter articulating with the truck instead of being free to turn with the semitrailer.

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(4) Provide adapter capable of transmitting the loads and forces in accordance with DOT FMCSR and as specified herein.

(d) Provide lower coupler air slider with in-cab release control:

(1) Slider travel range providing minimum of 42-inch, fifth wheel settings, forward of rearmost axle ("FW" of SAE J701);
(2) Slider spring-loaded, positive latching type;
(3) Air operated slider release control, air toggle protected from accidental driver operation by spring-loaded red plastic cover.

(e) Provide lower coupler and slider installation on chassis in accordance with DOT FMCSR and SAE publication SP-354 for T-section or double angle mount:

(1) All fasteners of high strength types:
Bolts per IFI STD 103 and 111, Grade 8;
Nuts of prevailing torque type per IFI STD 100, 101, 107, and 108, Grade C;
Integral flange fasteners or hardened steel washers under all nuts and bolts to prevent imbeding;
Minimum 5/8-inch diameter size, web fasteners located at least 1.75-inches from top of frame rails and spaced along frame at 6-inch or less intervals, the full length of the slider;
(2) Provide front and rear safety stop bars of the bolt-on type; Bars of minimum 1-inch thick by 2- x 3-inch size;
(3) Locate front bars at maximum forward slider travel and rear bars at center of drive tandem to comply with DOT FMCSR maximum slider fifth wheel setting;
(4) Provide for repositioning of rear bars to rearmost slider travel for special, off-road usage;
(5) Provide electrical ground cable between lower coupler plate and main frame, insuring metal-to-metal electrical path.

(f) Provide rear of frame bolt-on semitrailer pick-up ramps per TTMA RP No. 8 on each frame rail; provide replaceable wear plates; radius wear plate inner edges to preclude cutting or damage to semitrailer kingpin during offset coupling; bevel and box-in ends of frame rails.

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(g) Provide rear tail roller for winch cable and semitrailer nose roll-up:

(1) Minimum roller size frame-width wide and 8-inch diameter fabricated of steel equal to ASTM 120, Schedule 80 pipe having minimum 1-1/2-inch solid shaft through axle out each end; minimum 3/4-inch thick plate frame-to-axle longitudinal brackets bolted to frame rails;
(2) Roller located at rearmost part of truck tractor below the level of a semitrailer kingpin, when ascending the lower coupler frame ramps;
(3) Roller, brackets, and installation to frame withstand minimum of 45,000 pounds horizontal force in the forward direction and a minimum 45,000 pound vertical weight force downward.

(h) Provide light weight safety deck plates behind chassis-cab in accordance with SAE J185, and the following:

(1) Minimum 1/4-inch thick aluminum diamond safety plate, reinforced to prevent deformation.
(2) Minimum coverage from headache rack to just forward of slider guides and rear quarter fenders and width of at least 84-inches.
(3) Maximum 2-inch deck plate height above frame rails.
(4) Deck plates removable in multiple sections with each section secured to tractor by rubber hood latches, interchangeable with tilt hood.
(5) For serviced components, including applicable batteries and filters, provide hinged access covers.

3.2.6 Hydraulic system. The truck tractor shall be equipped with a hydraulic system for the winch and a double wet line for hydraulically operated semitrailer components. The hydraulic hoist system shall be as follows:

(a) Provide the complete hydraulic system having all components as recommended by the hydraulic pump and the winch manufacturer's recommendations.

(1) Operate with rated payload without binding, malfunction, or permanent deformation;
(2) Provide a fail-safe system protected from accidental winch disengagement due to any hydraulic system failure;

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(3) Capable of withstanding minimum proof pressures without leakage as recommended by the pump manufacturer;
(4) All components and lines clean and free of foreign matter;
(5) Hydraulic oil approved by the pump manufacturer for severe service.

(b) Provide a hydraulic winch pump, per SAE J744 of the positive displacement type having a minimum rating of 35 gpm capacity for a 2000 psi system:

(1) Provide pump integrally mounted on the PTO of the main transmission;
(2) Provide a pump having the pump manufacturer's continuous duty ratings for speed and performance equal to or in excess of the required speed and performance necessary for winch and semitrailer operation;
(3) Provide relief valve per SAE J747.

(c) Provide a hydraulic reservoir as follows:

(1) Capacity matched to hydraulic motor;
(2) Provide filler cap captive chained to a strainer in the filler neck, and marking "HYD RES" on or near reservoir cap;
(3) Provide self-sealing dipstick or sealed and guarded oil level indicator;
(4) Provide a reservoir air breather of the spin-on pleated paper canister type having a minimum size of 5-inches long by 3.7-inches diameter threaded on a SAE J363 3/4-16 UNF2A insert;
(5) Provide reservoir mounting to prevent damage from vibration and from off-road terrain.

(d) Provide hydraulic system filter in accordance with ANSI B93, as certified by the filter manufacturer, having the following:

(1) Minimum 6.2 filtration ratio at 20 micrometers and a dust capacity of a minimum of 1 gm per gpm flow;
(2) Provide minimum 100 psi housing burst pressure;
(3) Provide automatic by-pass valve;
(4) Provide a filter element service indicator gage.

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(e) Provide in-cab control valve(s) per SAE J747:

(1) Provide a winch lever control, accessible to the driver from his seated position;
(2) Provide control identification and position labeling on quadrant;
(3) Provide lever of the tower mount type having neutral "HOLD" lock of the hinged gate type preventing accidental engagement of the winch;
(4) Provide lever to valve flexible push-pull control cable of the waterproof, noncorrosive and stainless steel SAE J917 type;
(5) Provide control valve of the multi-position spring centered spool type having spool wiper and seals;
(6) Provide winch operation of winch-in, hold, and winch-out;
(7) When specified, (see 6.2), provide in-cab lever control(s) as specified for semitrailer hydraulically operated equipment functions.

(f) Provide flexible hydraulic hose lines between all components as follows:

(1) Provide hose per SAE J516 and SAE J517 having a working pressure rating in excess of the relief valve setting;
(2) Maintain hose bend sharpness in accordance with SAE J517 limits;
(3) Provide fittings at parts of the components in accordance with SAE J514, O-ring boss type, or SAE J518;
(4) Provide test point connection at pump outlet line for measuring pressure;
(5) Provide lines with reuseable fittings or provide replacement lines with reuseable fittings and bulk hose in the parts manual.

3.3 Chassis-cab. The truck tractor equipment shall be installed on a chassis-cab conforming to MIL-C-52895. The complete truck tractor shall be of the 8 x 4 type, including pusher axle, and shall be as follows:

(a) Weights and dimensions:

(1) GVWR - Minimum 69,000 pounds.
(2) GCWR - Minimum 120,000 pounds.

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(3)	Coupler Plate Weight (Payload)	- 47,000 pounds.
(4)	OAL	- Maximum 360-inches.
(5)	WB	- Per truck manufacturer's recommendations.
(6)	BBC, Conventional Cab.	- Range 114 through 126-inches.
(7)	BBC, Tilt Cab.	- Range 50 through 66-inches.
(8)	CA	- Range 140 through 165-inches.

(b) The truck tractor performance requirements shall be applicable at both GVWR and at GCWR, the latter while towing a semitrailer.

(c) The truck tractor performance of the lowest working speed shall be that required to achieve rated governed engine rpm without exceeding a road speed of 5.6 mph and without braking.

(d) The truck tractor main transmission driven PTO shall be tailored in speed and power to match the winch hydraulic system and provide simultaneous both winch-in and travel or winch-out and travel.

(e) The truck tractor shall be equipped with four axles, consisting of front axle, pusher axle, and rear tandem axles. The pusher axle control box shall be illuminated for night operation.

(f) The fender requirements for the rear triden shall be met by noncorrosive front quarter fenders, the full width goose neck of the semitrailer and rear splash shields on spring brackets which withstand impacts.

(g) The brake system shall include the following:

Hand control valve;
Tractor protection valve;
Trailer brake lines at the rear of the cab for truck tractors;
Auxiliary trailer brake lines at the rear of the frame for straight trucks.

(h) The electrical system shall include the following:

Trailer electrical cable for truck tractors;
Auxiliary trailer electrical receptacle at rear of the frame for straight trucks;
All rear lamps banked on a metal plate under the rear tail roller and on each side of the pintle.

(i) The truck tractor shall include the SAE J849 rear pintle.

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(j) For semitrailer kingpin clearance, the rear frame cross member shall be of the drop center or the inverted type.
(k) The truck tractor cab shall be as follows:

(1) Unless otherwise specified, the cab shall be of the conventional CAE type;
(2) When specified (see 6.2) the cab shall be of the cab-over-engine (COE), tilt-cab type as follows:

Lightweight type of aluminum or fiberglass reinforced plastic and aluminum material;
Minimum 70 degree tilt or two stage of minimum 50 degree and 70 degree tilt providing walk-up engine access;
Power tilt system of electrohydraulic or air-hydraulic types with manual hydraulic emergency pump;
Safety tilt cab locking system;
Shock and vibration reduction cab mounts;
Minimum 81-inch width by minimum 54-inch height interior cab size.

(l) The truck tractor shall be equipped with a 7-day recording speedometer and tachometer in lieu of the nonrecording types.

3.4 Lubrication. Lubrication means shall be provided for moving parts that require lubrication. The types of lubricants shall be selected from and in accordance with SAE J754 and, whenever practical, shall be interchangeable with military lubricants (see 6.4). All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Lubrication intervals shall be in accordance with SAE J752. Grease lube fittings shall conform to SAE J534 and shall be of the threaded types. Fluid level indicators shall be in accordance with SAE J48 and SAE J614. A lubrication chart in accordance with SAE J753 shall be furnished and shall include the interchangeable military lubricants as well as the commercial SAE J754 lubricant designations.

3.5 Maintenance characteristics. The truck tractor design shall permit ready access to all items requiring periodic maintenance service in the field. Periodic maintenance shall be accomplished with the use of conventional general purpose tools normally associated with equipment of this type. The adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other components of the truck tractor.

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3.6 Slinging and tiedowns. Procedures for lifting and tiedown of the complete truck tractor shall be provided. These procedures shall be included in the appropriate manuals.

3.7 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded, or insulated to the extent necessary to eliminate the hazard. External equipment including platforms, handrails, fenders and bumpers shall be designed for durability and operator safety. All steps shall be provided with anti-skid surfacing. All platforms, steps, and handholds shall be of such design to permit unhampered and nonhazardous access to and from the truck. If the noise level in the area occupied by the operator exceeds 85db(A), as measured in accordance with SAE J919, a warning plate containing the following shall be permanently affixed to the mixer in a conspicuous protected location:

"Warning, Ear Protection Required When This Truck Tractor Is Operating."

The plate coloring shall be yellow and black as specified in the chassis for ANSI warning plate requirements.

3.8 Treatment and painting. The parts normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice. The final color shall be forest green in accordance with MIL-T-704. The inaccessible areas of the truck tractor equipment shall be treated with anti-rust coating in accordance with TTMA Technical Bulletin No. 56.

3.9 Marking. The U.S. Army marking and registration numbers shall be as specified in the solicitation. Marking and registration shall be in accordance with MIL-STD-642, color black. The GVWR and GCWR shall be marked on the lower bottom of the cab doors.

3.10 Government-furnished equipment (GFE).

(a) Case, Maintenance and Operator's Manuals, NSN 7520-00-559-9618.
(b) Materiel Fielding Plan.
(c) Prescribed Load List (PLL) and Authorized Stock List (ASL).
(d) Vehicle classification sign kit (reference MIL-S-40626, NSN 9905-00-565-6267).

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3.11 Basic issue items and commercial manuals. Each truck tractor shall be overpacked with:

(a) The items specified in Basic Issue Items List.
(b) Two copies of the Government-approved Operator, Parts, Maintenance and Overhaul Manuals, and Lubrication Order.

3.12 Identification and instruction plates. The manufacturer shall provide all identification and instruction plates normally furnished. The plates and their installation shall conform to the "Metal Name Plate Industry Standards and Practices", published by NAME. The plates shall be permanently affixed to the truck tractor in a conspicuous location.

3.13 Mounting of GFE. The manufacturer shall provide fasteners and mount the GFE listed in 3.10.(a).

3.14 Questionnaire for Evaluation. The contractor shall complete and submit a Questionnaire for Evaluation within the time frame specified (see 6.2 and 6.5). The information shall depict the complete truck tractor and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used.

3.15 Vehicle classification. The manufacturer shall complete and submit the Government-furnished "Vehicle Classification Data Sheet" at the time of submission of his proposal. The Government-furnished sign kit shall be installed by the contractor on the front of the truck in a location approved by the contracting officer.

3.16 Technical publications. Such technical publications as are specified shall be furnished (see 6.2).

3.17 Special tools. Any tools required for operation and operator maintenance shall be provided with each truck tractor.

3.18 Certification. Prior to the first article evaluation, (see 4.2), the manufacturer shall provide certification that the truck tractor conforms to the following to the extent specified herein:

(a) DOT FMVSS.
(b) DOT FMCSR.

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(c) California Vehicle Code.
(d) DOL OSHA Regulations.
(e) EPA Regulations.
(f) Exterior Noise Level.
(g) Interior Noise Level.
(h) Radio Suppression.
(i) Industry Standardization Document Compliance.
(j) Applicable GVWR - GCWR Performance.

3.19 Transportability. The truck tractor shall be transportable with operational equipment up to GCWR.

3.19.1 Aircraft cargo. The truck tractor shall be transportable in a C-5A aircraft as shown in Figure 1.

3.19.2 Railroad loading. The truck tractor shall be rail transportable on a flatcar to comply with the requirements of the Rules Governing the Loading of Commodities on Open Top Cars of the Association of American Railroads.

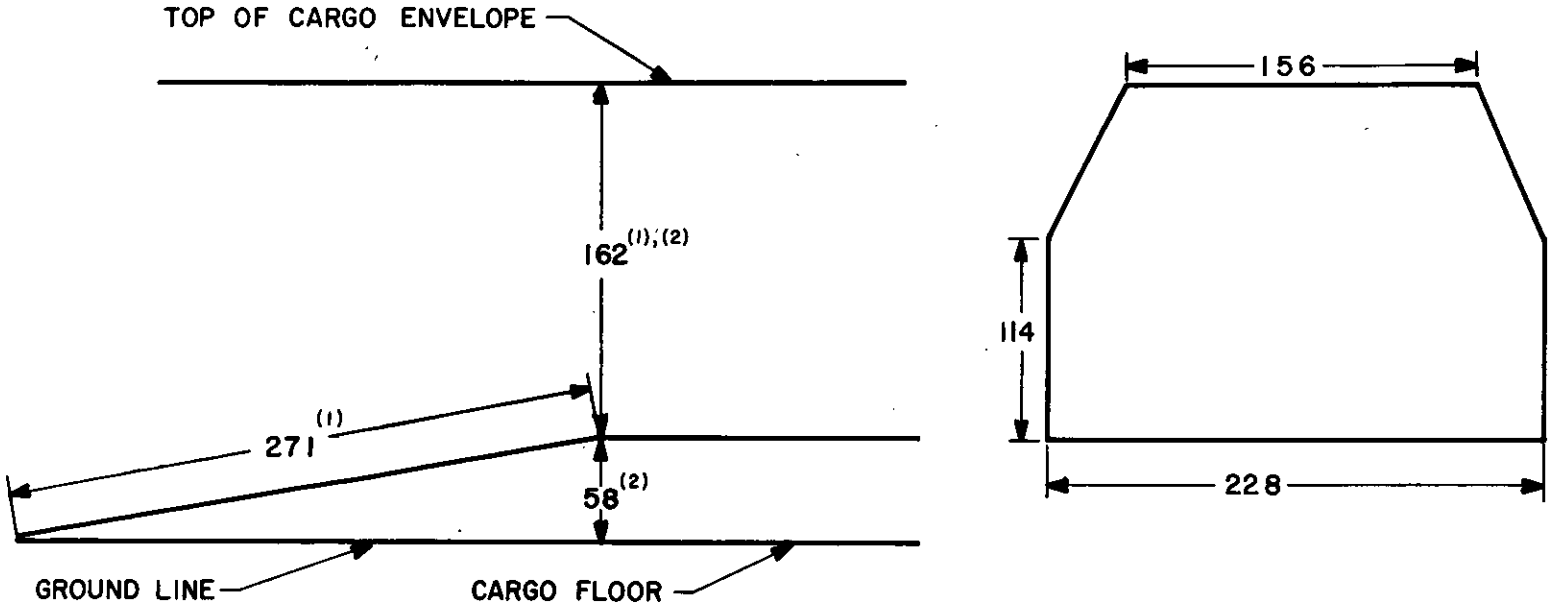
4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article evaluation. The first-produced truck tractor, with its components, shall be compared with this document, MIL-C-52895(ME), the manufacturer's completed Questionnaire for Evaluation, published literature, calculations and test data, bill of Materials and Assembly Line Build-up Sheet, to verify conformance to this specification. In addition, the truck tractor shall be subjected to a demonstration as specified herein. Nonconformance to any requirement contained herein shall be cause for rejection of the truck tractor.

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FORWARD CARGO OPENING C-5A
FORWARD KNEEL POSITION



NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. REFERENCE: AMCP 706-130, ENGINEERING DESIGN HANDBOOK, DECEMBER 1967.
3. REFERENCE: USATECOM PROJECT NO 4-ES-065-050-004, SERVICE TEST OF C-5A AIRCRAFT, FIRST PARTIAL REPORT BY H.H JOHNSON, JR., CPT., JANUARY 1971.

FIGURE 2. C-5A. cargo envelope.

X-3171

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4.2.1 Performance. Demonstrate the ability of the truck tractor to perform the following:

(a) Perform semitrailer coupling, towing, and uncoupling cycles for various available semitrailers as follows:

Long types of at least 40 foot length;
Tanker types;
Heavy hauler (low bed) machinery types;
Short types of 24 foot length or less.

4.2.2 Truck tractor weights. Determine the following truck tractor weight by weighing and recording the results, under the following conditions:

(a) Determine weights both axle-by-axle and total with the pusher axle both load carrying and in the raised position:

(1) Truck tractor's curb weight;
(2) Truck tractor's actual GVW with a lower coupler plate payload of 47,000 pounds for SAE J701 slider FW settings of at least 26-, 30-, 36-, 46-, 56-, and the maximum travel from the rearmost axle; tabulate or graph results;
(3) Truck tractors actual lower coupler plate payload at GVWR and the optimum FW setting of the slider for GVWR.

(b) With the pusher axle raised, determine the truck tractor's recommended maximum speeds allowed by the TRA Yearbook with a lower coupler plate payload of 47,000 pounds at the raised pusher optimum FW setting.
(c) Compare the actual pusher axle air bag pressures and axle weights with the load-pressure chart furnished.

Compare actual weights with the component ratings and the vehicle ratings. In the 8 x 4 configuration, actual weights in excess of the DOT GAWR, GVWR, and GCWR shall constitute failure of this test. Weight on any axle in excess of 18,000 pounds shall constitute failure of this demonstration.

4.2.3 Highway speed. Demonstrate the truck's maximum high gear, highway speed capability on hard surfaced roadway with the vehicle loaded to GCWR as specified in 3.3. Average the speed for four runs, two in each direction. Inability to operate at 50 miles per hour shall constitute failure of this demonstration.

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4.2.4 Gradeability. Demonstrate the gradeability of the truck tractor and semitrailer when loaded to GCWR as specified in 3.3. Calculate and demonstrate driving the truck tractor up not less than a 20% grade, stop, and park the truck, restart and continue ascending the slope. Calculate and demonstrate the ability of the truck tractor when loaded to GCWR as specified in 3.3, to ascend a 3% grade at not less than 20 mph. Calculations shall be in accordance with SAE J688 and demonstrations shall be in accordance with SAE J950. Inability to ascend, stop and restart on the 20% grade or inability to ascend a 3% grade at 20 mph shall constitute failure of this demonstration.

4.2.5 Truck tractor mobility. The truck tractor mobility shall be performed as required herein. Permanent deformation, overheating, malfunction, failure to complete any test or leakage of fuel, lubricants or hydraulic fluid shall be cause for rejection of the first article.

Terrain	Speed	Loading
Primary Highway Types	50 miles @ up to max. mph	GCWR
Primary Highway Types	50 miles @ up to max. mph	Empty
Secondary Roads	25 miles @ up to 60% max. mph	GCWR
		Empty
Secondary Roads	25 miles @ up to 60% max. mph	
Off-Road Terrain	10 miles @ safe speeds	GCWR
Off-Road Terrain	5 miles @ safe speeds	Empty

During the operation, demonstrate the following:

(a) Ability to load, transport, and unload.
(b) Based on the primary highway fuel consumption both loaded and empty, calculate travel distance per tank full.
(c) Rocking of the truck tractor with load in soft, slippery, off-road terrain by cycle shifting at not more than 2-second intervals from forward to reverse and back, continuously for at least 60 seconds. Achieve at least 3/4 wide-open-throttle power in each gear. Repeat the cycle rocking without load (bob-tailed).

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(d) Apply the service brakes, shift the transmission into lowest forward gear, and apply wide-open throttle power for not less than 5 seconds (or until the engine stalls). Repeat in all gears, including reverse.

4.2.6 Turning capability. Demonstrate vehicle clearance circle and curb clearance circle in accordance with SAE J695, Field Test Prodecures.

4.2.7 Braking capability. Demonstrate the brake capability of the truck. Conduct service, emergency, and parking ability demonstrations in accordance with DOT and California Vehicle Code requirements. Inability to complete this test shall constitute failure of this demonstration.

4.2.8 Cooling system data. The contractor shall supply certified test data or detailed calculations for the truck tractor to prove that sufficient cooling system is furnished to maintain a radiator top tank temperature not greater than 212°F at an ambient temperature of 120°F. The engine oil, convertor, and transmission fluid (during lock-up operation) and power steering system hydraulic oil shall not be greater than 250°F, the convertor outlet temperature shall not exceed 300°F during convertor operation as required to start and climb the specified grade not less than 75 feet in length, with the maximum overall ratio reduction, at an ambient temperature of 120°F. Either test data or calculations are acceptable.

4.2.9 Certification. The contractor shall be responsible for insuring that components and materials used in the truck tractor are manufactured, examined, and tested in accordance with the requirements and references specified herein. As a minimum, the contractor and his component subcontractors through the truck contractor shall be required to certify the following:

(a) In effect at actual time of manufacture:

(1) DOT FMVSS.
(2) DOT FMCSR.
(3) California Vehicle Code.
(4) DOL OSHA Regulations.

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(5) EPA Regulations.
(6) Exterior Noise Level.
(7) Interior Noise Level.
(8) Radio Suppression.
(9) Transportability (see 3.19).

(b) In effect at time of solicitation for bid:

(1) Industry Standardization Document Compliance.
(2) Applicable GVWR - GCWR chassis performance.
(3) Ability to withstand water crossing operation (see MIL-C-52895(ME)).

4.3 Run-in. Each truck tractor shall be subjected to a run-in and servicing as specified herein. Inability to complete the run-in, malfunction of any control, evidence of leakage of fuel, lubricants or hydraulic fluid shall be cause for rejection of the truck tractor by the Government until such deficiencies are corrected.

4.3.1 Tractor equipment portion. Each component shall be run-in, lubricated and calibrated in accordance with the component manufacturer's current procedures.

4.3.2 Truck chassis.

4.3.2.1 Drive train. Apply the service brakes, shift the transmission into gear, and apply wide open throttle for not less than 5 seconds. Repeat in all gears including reverse.

4.3.2.2 Travel. Demonstrate starting, shifting up and down through all gears, braking, steering, and high speed travel for a distance of at least 2 miles.

4.3.2.3 Servicing and adjustment. Prior to delivery, the manufacturer shall service and adjust each truck for customer operational use including at least the following:

(a) Alinement of steering and the front end assembly.
(b) Adjustment of engine, transmission(s), and power train.
(c) Adjustment of the brake system.

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(d) Continuity inspection of the electrical system, including filling and charging of the batteries.
(e) Check torque of all wheel lug nuts and inflation of all tires.
(f) Servicing of the cooling system with antifreeze solution, 50-50 percent by volume.

4.4 First article demonstration report. The contractor shall prepare and submit the first article demonstration report to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination and demonstration performed, deficiencies encountered and actions taken to correct the deficiencies.

5. PACKAGING

5.1 Preservation and packing. The truck tractor and all components and attachments shall be preserved and packed for domestic or overseas shipment as specified (see 6.2), in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The truck tractors covered by this specification are intended for world-wide earthmoving, construction, and materials handling equipment hauling operations by the Armed Forces in engineer troop support use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When an in-cab level control valve for a folding goose neck raise and lower hydraulic cylinder system is required (see 3.2.6(e)(7).
(c) When a tilt cab is required for the truck tractor (see 3.3(k)(2).
(d) U.S. Army marking and registration numbers (see 3.9).
(e) Items listed in the Basic Issue Item List (see 3.11).
(f) Time frame for completing and submitting Questionnaire (see 3.14 and 6.4).

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(g) Time frame for submitting the first article demonstration report (see 4.4).
(h) Technical publications to be supplied.
(i) Type of packaging required (see 5.).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.10.

6.4 Publications. The contracting officer will determine that the manufacturer's operating, maintenance, and parts manual and lubrications instructions are approved by the Government prior to accepting the first article.

6.5 Questionnaire for Evaluation. The Questionnaire for Evaluation may be obtained from MERADCOM, ATTN: DRXFB-HK, Fort Belvoir, VA 22060 (see 3.14).

Custodian:

Army - ME

Preparing activity:

Army - ME

Project 3805-A092

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Commander
US Army Mobility Equipment Research and
Development Command
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Fort Belvoir, VA 22060

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-52901(ME) TRUCK TRACTOR, ON-OFF ROAD, ENGINEER MIN. 120,000 GCWR, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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